Project Phase 3: Relational Database Schema — Towerrunning World Database (TWDB)

> **Domain & Phase 1 Requirements:** Team 63 (Carromboard) — *Towerrunning World Database (https://www.towerrunning.com)*
> **Assigned Phase 2 EER Model:** Team 59 (Aggregays) — *13 Entity Types, 15 Relationship Types, 110 Attributes*
> **Interactive draw.io Relational Schema Diagram Link:** Click Here to Open Relational Schema Diagram in draw.io (diagrams.net) | *Editable File:* `TWDB_Relational_Schema.drawio`

1. Overview of the EER-to-Relational Mapping

We convert Team 59's Enhanced Entity-Relationship (EER) diagram into a relational database schema following **Elmasri & Navathe Chapter 9 (Steps 1–9)** and **Items 1–8 of the Phase 3 instructions**. As instructed, **no normalization** is performed beyond the direct mapping rules, and all 16 derived attributes are omitted from base storage and documented below.

The resulting relational schema contains **19 relations** and **24 foreign key constraints** (25 arrows in the diagram, as the composite foreign key to `COURSE` spans 2 columns): **9 strong entity relations** (`NATION`, `NATIONAL_ASSOCIATION`, `PERSON`, `VENUE`, `SEASON`, `CHAMPIONSHIP`, `RACE`, `RANKING_EDITION`, `SCORING_RULE`), **2 weak entity relations** (`COURSE`, `MEMBERSHIP`), **2 subclass relations** (`OFFICIAL`, `REGISTERED_ATHLETE`), **2 multi-valued attribute relations** (`VENUE_PHOTO_URLS`, `RACE_MEDIA_LINKS`), **3 binary M:N relationship relations** (`RANKED_IN`, `NATION_RANKED_IN`, `APPLIES_FOR`), and **1 ternary (n = 3) relationship relation** (`FINISHED`).

2. Step-by-Step Mapping Explanation (Following Instructions 1–8)

2.1 Strong Entity Types, Composite Attributes, and Derived Attributes (Rule 1 / Step 1)

- **`NATION`**: Mapped with simple attributes `Nation_Code`, `Nation_Name`, and `Continent`. Both `Nation_Code` and `Nation_Name` are candidate keys in the EER diagram; we choose **Nation_Code** as the primary key and keep `Nation_Name` as a `UNIQUE` alternate key.
- **`NATIONAL_ASSOCIATION`**: Mapped with primary key **Assoc_ID** and simple attributes `Assoc_Name` and `Founded_Year` (plus `Contact_Link_1..3` from Rule 4(a) and foreign key `Nation_Code` from `GOVERNS`).
- **`PERSON`**: Mapped with primary key **Person_ID**, simple attributes `Gender` and `Date_of_Birth`, and foreign key `Nation_Code` (from `CITIZEN_OF`).
  - *Composite attribute flattened:* `Name` becomes simple columns `First_Name` and `Last_Name`.
  - *Derived attribute not stored:* `Age` is omitted because it is computed dynamically from `Date_of_Birth` and the current date.
- **`VENUE`**: Mapped with primary key **Venue_ID**, simple attributes `Venue_Name`, `Venue_Type`, `Building_Height_m`, and foreign key `Nation_Code` (from `LOCATED_IN`).
  - *Composite attribute flattened:* `Address` becomes simple columns `Street`, `City`, and `Postal_Code`.
  - *Derived attributes not stored:* `Course_Record_Male` and `Course_Record_Female` are omitted because they are computed as the fastest evaluated `FIN` time per gender across courses at that venue.
- **`SEASON`**: Mapped with primary key **Season_Year**, simple attributes `Start_Date`, `End_Date`, `Invites_Per_Gender`, `Wildcards_Per_Gender`, `Invitation_Cutoff`, and foreign key `Final_Race_Code` (from `FINAL_OF`).
- **`CHAMPIONSHIP`**: Mapped with primary key **Championship_ID** and simple attributes `Title`, `Level`, `Region`, and `Year`.
- **`RACE`**: Mapped with primary key **Race_Code**, simple attributes `Race_Name`, `Race_Date`, `Duration_Days`, `Edition_No`, `Factor`, `Status`, `Evaluation_Status`, `Results_Received_On`, and `Additional_Info` (plus foreign keys/attributes from `HELD_AT`, `PART_OF`, `DECIDED_AT`, and `PRECEDED_BY`).
  - *Composite attribute flattened:* `Contact` becomes simple columns `Contact_Person` and `Contact_Email`.
  - *Derived attributes not stored:* `Category` (derived from whether `Factor >= 60`), `Male_Winner`, and `Female_Winner` (derived from `FINISHED` where `Gender_Place = 1` on the evaluated course) are omitted.
- **`RANKING_EDITION`**: Mapped with primary key **Ranking_ID** and simple attributes `Edition_Label`, `Period_Start`, `Period_End`, `Revision_No`, `Is_Season_Final`, and `Published_On`.

- **SCORING_RULE**: Mapped with simple attribute Points. The composite key Factor_Place is flattened into its simple components **Factor** and **Place**, forming the composite primary key (Factor, Place).

2.2 Weak Entity Types and Identifying Relationships (Rule 2 / Step 2)

- **COURSE (Owner: RACE via identifying relationship R4 HAS_COURSE)**: Includes owner primary key **Race_Code** as a foreign key referencing RACE(Race_Code) combined with partial key **Course_Name** to form composite primary key (Race_Code, Course_Name) (absorbing R4 HAS_COURSE). Includes simple attributes Race_Mode, Upward_Stairs, Downward_Stairs, Floors, Vertical_Metres, Flat_Section_m, Special_Info, and Stair_Time_Share_Pct.
  - *Derived attribute not stored:* Is_Evaluated is omitted (derived as true for the course with maximum Upward_Stairs in its race).
- **MEMBERSHIP (Owner: REGISTERED_ATHLETE via identifying relationship R5 HOLDS)**: Includes owner primary key **Athlete_ID** as a foreign key referencing REGISTERED_ATHLETE(Athlete_ID) combined with partial key **Member_Year** to form composite primary key (Athlete_ID, Member_Year) (absorbing R5 HOLDS).
  - *Composite attribute flattened:* Payment becomes Payment_Method and Payment_Reference, alongside simple attributes Paid_On and Status.
  - *Derived attribute not stored:* Fee_EUR is omitted (derived from Member_Year and birth year: €18 if age ≤ 20 or ≥ 50, else €24).

2.3 Binary 1:1, 1:N, and M:N Relationships (Rule 3 / Steps 3, 4, 5)

- **Binary 1:1 Relationships (Step 3 — Foreign Key Approach):**
  - R9 GOVERNS (NATIONAL_ASSOCIATION (1,1) — NATION (0,1)): Foreign key Nation_Code (UNIQUE NOT NULL, referencing NATION(Nation_Code)) placed in NATIONAL_ASSOCIATION because NATIONAL_ASSOCIATION has total participation (1,1).
  - R7 FINAL_OF (SEASON (0,1) — RACE (0,1)): Both sides are partial; foreign key Final_Race_Code (UNIQUE, nullable, referencing RACE(Race_Code)) placed in SEASON (1 row per year, avoiding NULLs across hundreds of non-final RACE rows).
- **Binary 1:N Relationships (Step 4 — Foreign Key on the N-Side):**
  - R1 CITIZEN_OF: Foreign key Nation_Code in N-side relation PERSON referencing NATION(Nation_Code).
  - R2 LOCATED_IN: Foreign key Nation_Code in N-side relation VENUE referencing NATION(Nation_Code).
  - R3 HELD_AT: Foreign key Venue_ID in N-side relation RACE referencing VENUE(Venue_ID).
  - R6 PART_OF: Foreign key Season_Year (referencing SEASON(Season_Year)) and relationship attribute Season_Race_No placed in N-side relation RACE.
  - R8 DECIDED_AT: Foreign key Championship_ID (nullable, referencing CHAMPIONSHIP(Championship_ID)) placed in N-side relation RACE (each race decides at most 1 championship (0,1)).
  - R10 REPRESENTS: Foreign key Assoc_ID (nullable, referencing NATIONAL_ASSOCIATION(Assoc_ID)) placed in N-side relation OFFICIAL (each official represents at most 1 association (0,1)).
- **Binary M:N Relationships (Step 5 — Separate Relationship Relations):**
  - **RANKED_IN (R11)**: Composite primary key (**Athlete_ID**, **Ranking_ID**) referencing REGISTERED_ATHLETE and RANKING_EDITION, with stored attributes Rank_Position, Score_Points, Regular_Races, Tour_Races. *Derived attributes not stored:* Total_Races, Rank_Change, and Score_Change.
  - **NATION_RANKED_IN (R12)**: Composite primary key (**Nation_Code**, **Ranking_ID**) referencing NATION and RANKING_EDITION, with stored attributes Rank_Position, Score_Points, Regular_Races, Tour_Races. *Derived attributes not stored:* Total_Races, Rank_Change, and Score_Change.
  - **APPLIES_FOR (R13)**: Composite primary key (**Athlete_ID**, **Race_Code**) referencing REGISTERED_ATHLETE and RACE, with stored attributes Applied_On, Application_Status, and Travel_Reimbursement_EUR.

2.4 Multi-Valued Attributes (Rule 4 / Step 6)

1. **Contact_Links on NATIONAL_ASSOCIATION — Option 4(a) (Same relation, Maximum = 3 values: Contact_Link_1, Contact_Link_2, Contact_Link_3):** *One-line reason:* Sections 1.1 and 2.1 of the requirements specify a small, fixed maximum of three contact links per national association (website, Facebook, and e-mail).
2. **Photo_URLs on VENUE — Option 4(b) (Separate relation VENUE_PHOTO_URLS(Venue_ID, Photo_URL), FK Venue_ID → VENUE):** *One-line reason:* A venue hosts many race editions across years and can accumulate an unbounded number of building photographs over time.

3. **`Media_Links` on `RACE` — Option 4(b) (Separate relation `RACE_MEDIA_LINKS(Race_Code, Media_Link)`, FK `Race_Code` → `RACE`):** *One-line reason:* A race can have an unbounded number of media URLs (race link, result link, TWA announcement, and multiple videos and photo galleries).

2.5 Relationships of Degree n > 2 (Rule 5 / Step 7)

- **`FINISHED` (`R15` — Ternary relationship among `PERSON`, `COURSE`, and `NATION`)**: Mapped to relation `FINISHED` with foreign keys `Person_ID` → `PERSON(Person_ID)`, `(Race_Code, Course_Name)` → `COURSE(Race_Code, Course_Name)`, and `Nation_Code` → `NATION(Nation_Code)`, plus simple attributes `Finish_Time` and `Result_Status`. Its primary key is `(Person_ID, Race_Code, Course_Name)` (excluding `Nation_Code` per Step 7 because each person finishes a given course at most once for a single race-day nation). *Derived attributes not stored:* `Gender_Place` and `Points`.

2.6 Subclasses and Superclasses — Specialization Mapping (Rule 6 / Step 8)

The specialization of `PERSON` into `OFFICIAL` and `REGISTERED_ATHLETE` is **overlapping (`o`) and partial**. We used **Textbook Option 8A (Multiple relations — superclass and subclasses)** because Option 8B fails on partial/overlapping specializations and Options 8C/8D would create sparse `NULL` columns on `PERSON` while complicating subclass-specific relationships (`REPRESENTS`, `HOLDS`, `RANKED_IN`, `APPLIES_FOR`):

- **`OFFICIAL`**: Primary key `Person_ID` (foreign key referencing `PERSON(Person_ID)`), local attributes `Official_Role` and `Since_Year`, and foreign key `Assoc_ID` (from `REPRESENTS`).
- **`REGISTERED_ATHLETE`**: Primary key `Athlete_ID`, foreign key `Person_ID` (`UNIQUE NOT NULL`, referencing `PERSON(Person_ID)`), `UNIQUE` alternate keys `Username` and `Email`, and local attribute `Registered_On`. *Derived attribute not stored:* `Current_Score` (sum of best 8 race points in the last 12 months).

2.7 Recursive Relationships (Rule 7 / Step 3)

- **`R14 PRECEDED_BY` (`RACE as next (0,1)` — `RACE as previous (0,1)`, recursive 1:1)**: Mapped using self-referencing foreign key `Prev_Race_Code` (`UNIQUE`, nullable) in `RACE` referencing `RACE(Race_Code)`, pointing from a race edition to its immediately preceding edition.

2.8 Union Types / Categories (Rule 8 / Step 9)

- No union types (categories) or shared subclasses exist in the EER diagram, so Step 9 is not needed.

3. Minimal Corrections / Gaps Noted in the EER Diagram

1. **Subclass Key of `REGISTERED_ATHLETE` (Phase 2 Finding M3):** In Team 59's EER diagram, `REGISTERED_ATHLETE` underlines `Athlete_ID`, `Username`, and `Email` besides inheriting `Person_ID`, and weak entity `MEMBERSHIP` uses `Athlete_ID` in its full key `(Athlete_ID, Member_Year)`. Under Option 8A, we store `Person_ID` as a `UNIQUE NOT NULL` foreign key referencing `PERSON(Person_ID)` and designate `Athlete_ID` as the primary key of `REGISTERED_ATHLETE` so that `MEMBERSHIP`, `RANKED_IN`, and `APPLIES_FOR` reference `Athlete_ID` directly.
2. **Ternary `FINISHED` Key Constraint (Phase 2 Finding C2 & M2):** Because `(min,max)` notation on a ternary cannot express `(PERSON, COURSE) → NATION`, the EER diagram labels all three edges `(0,N)`. Per Chapter 9 Step 7, we make `(Person_ID, Race_Code, Course_Name)` the primary key of `FINISHED` and keep `Nation_Code` as a non-key foreign key referencing `NATION(Nation_Code)`.

4. Complete Relational Schema Summary (All 19 Relations)

Every relation is listed below with all stored attributes, its **underlined primary key**, and all **foreign keys** (→ target primary key):

#	Relation Name	Attributes (Primary Key Underlined)	Foreign Keys (FK → Target Relation(PK))
1	**NATION**	Nation_Code, Nation_Name, Continent	*None* (Nation_Name UNIQUE)
2	**NATIONAL_ASSOCIATION**	Assoc_ID, Assoc_Name, Founded_Year, Contact_Link_1, Contact_Link_2, Contact_Link_3, Nation_Code	Nation_Code → NATION(Nation_Code)
3	**PERSON**	Person_ID, First_Name, Last_Name, Gender, Date_of_Birth, Nation_Code	Nation_Code → NATION(Nation_Code)
4	**OFFICIAL**	Person_ID, Official_Role, Since_Year, Assoc_ID	Person_ID → PERSON(Person_ID) Assoc_ID → NATIONAL_ASSOCIATION(Assoc_ID)
5	**REGISTERED_ATHLETE**	Athlete_ID, Person_ID, Username, Email, Registered_On	Person_ID → PERSON(Person_ID)
6	**MEMBERSHIP**	Athlete_ID, Member_Year, Payment_Method, Payment_Reference, Paid_On, Status	Athlete_ID → REGISTERED_ATHLETE(Athlete_ID)
7	**VENUE**	Venue_ID, Venue_Name, Venue_Type, Street, City, Postal_Code, Building_Height_m, Nation_Code	Nation_Code → NATION(Nation_Code)
8	**VENUE_PHOTO_URLS**	Venue_ID, Photo_URL	Venue_ID → VENUE(Venue_ID)
9	**SEASON**	Season_Year, Start_Date, End_Date, Invites_Per_Gender, Wildcards_Per_Gender, Invitation_Cutoff, Final_Race_Code	Final_Race_Code → RACE(Race_Code)
10	**CHAMPIONSHIP**	Championship_ID, Title, Level, Region, Year	*None*
11	**RACE**	Race_Code, Race_Name, Race_Date, Duration_Days, Edition_No, Factor, Status, Evaluation_Status, Results_Received_On, Additional_Info, Contact_Person, Contact_Email, Venue_ID, Season_Year, Season_Race_No, Championship_ID, Prev_Race_Code	Venue_ID → VENUE(Venue_ID) Season_Year → SEASON(Season_Year) Championship_ID → CHAMPIONSHIP(Championship_ID) Prev_Race_Code → RACE(Race_Code)
12	**RACE_MEDIA_LINKS**	Race_Code, Media_Link	Race_Code → RACE(Race_Code)
13	**COURSE**	Race_Code, Course_Name, Race_Mode, Upward_Stairs, Downward_Stairs, Floors, Vertical_Metres, Flat_Section_m, Special_Info, Stair_Time_Share_Pct	Race_Code → RACE(Race_Code)
14	**FINISHED**	Person_ID, Race_Code, Course_Name, Nation_Code, Finish_Time, Result_Status	Person_ID → PERSON(Person_ID) (Race_Code, Course_Name) → COURSE(Race_Code, Course_Name) Nation_Code → NATION(Nation_Code)
15	**APPLIES_FOR**	Athlete_ID, Race_Code, Applied_On, Application_Status, Travel_Reimbursement_EUR	Athlete_ID → REGISTERED_ATHLETE(Athlete_ID) Race_Code → RACE(Race_Code)
16	**RANKING_EDITION**	Ranking_ID, Edition_Label, Period_Start, Period_End, Revision_No, Is_Season_Final, Published_On	*None*

#	Relation Name	Attributes (Primary Key Underlined)	Foreign Keys (FK → Target Relation(PK))
17	**RANKED_IN**	Athlete_ID, Ranking_ID, Rank_Position, Score_Points, Regular_Races, Tour_Races	Athlete_ID → REGISTERED_ATHLETE(Athlete_ID) Ranking_ID → RANKING_EDITION(Ranking_ID)
18	**NATION_RANKED_IN**	Nation_Code, Ranking_ID, Rank_Position, Score_Points, Regular_Races, Tour_Races	Nation_Code → NATION(Nation_Code) Ranking_ID → RANKING_EDITION(Ranking_ID)
19	**SCORING_RULE**	Factor, Place, Points	*None*

Relational Schema Diagram in draw.io: Open Interactive Diagram in draw.io (diagrams.net) (Also provided as editable source `TWDB_Relational_Schema.drawio`, standalone full-bleed vector sheet `TWDB_Relational_Schema_Diagram.pdf`, and embedded in full landscape on the next page.)

5. Relational Database Schema Diagram (Elmasri & Navathe Figure 9.2 Notation)

